IEOR 221 - HW 7 - Jie Wen

1

(a)

Under Black–Scholes–Merton with no dividends, in the risk-neutral measure $\mathbb{Q}$,

$$\ln S_T \mid \mathcal{F}_t \sim \mathcal{N}\left(\ln S + \left(r - \tfrac{1}{2}\sigma^2\right)\tau, \sigma^2\tau\right), \quad \tau = T - t,$$

so

$$\mathbb{E}_{\mathbb{Q}}[\ln S_T \mid \mathcal{F}_t] = \ln S + \left(r - \tfrac{1}{2}\sigma^2\right)\tau.$$

Pricing is discounted risk-neutral expectation:

$$f(S,t) = e^{-r\tau}\mathbb{E}_{\mathbb{Q}}[\ln S_T \mid \mathcal{F}_t] = e^{-r\tau}\left[\ln S + \left(r - \tfrac{1}{2}\sigma^2\right)\tau\right].$$

(Geometric Brownian motion under $\mathbb{Q}$ has drift r, giving the above normal law; by definition of the lognormal, the mean of $\ln S_T$ is that normal's mean. Risk-neutral pricing uses the discounted expectation.)

(b)

Let $A(t) = e^{-r\tau}$. Then

$$f(S,t) = A\left[\ln S + \left(r - \tfrac{1}{2}\sigma^2\right)\tau\right].$$

Derivatives:

$$\frac{\partial f}{\partial S} = \frac{A}{S}, \qquad \frac{\partial^2 f}{\partial S^2} = -\frac{A}{S^2},$$

and using $\frac{d\tau}{dt} = -1$ and $\frac{dA}{dt} = rA$,

$$\frac{\partial f}{\partial t} = rA\left[\ln S + \left(r - \tfrac{1}{2}\sigma^2\right)\tau\right] - A\left(r - \tfrac{1}{2}\sigma^2\right).$$

Plug into the Black–Scholes–Merton PDE (no dividends):

$$\frac{\partial f}{\partial t} + rS\frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2\frac{\partial^2 f}{\partial S^2} - rf = 0,$$

and observe term-by-term cancellation:

$$\left[rA(\cdots) - A\left(r - \frac{1}{2}\sigma^2\right)\right] + rA - \frac{1}{2}\sigma^2 A - rA(\cdots) = 0.$$

Hence f satisfies the BSM PDE.

Final answer:

$$f(S,t) = e^{-r(T-t)}\left[\ln S + \left(r - \tfrac{1}{2}\sigma^2\right)(T-t)\right],$$

and it satisfies the Black–Scholes–Merton equation.

2

- **1 day (1/250 yr):** $0.18\sqrt{\frac{1}{250}} \approx 0.01138 \Rightarrow 1.14\%$
- **1 week (1/52 yr):** $0.18\sqrt{\frac{1}{52}} \approx 0.02496 \Rightarrow 2.50\%$
- **1 month (≈1/12 yr):** $0.18\sqrt{\frac{1}{12}} \approx 0.05196 \Rightarrow 5.20\%$

3

Given: $S_0 = 1000, K = 950, r = 4\%$ (cont.), $T = 0.25$ yr, call $c = 78$, put $p = 26$.

(a)

Put–call parity with a continuous dividend yield:

$$c - p = S_0 e^{-qT} - K e^{-rT}.$$

Solve for e^{-qT} then q:

$$e^{-qT} = \frac{c - p + K e^{-rT}}{S_0} = \frac{78 - 26 + 950 e^{-0.04 \cdot 0.25}}{1000} \approx 0.992547,$$

$$q = -\frac{1}{T}\ln(0.992547) \approx 2.99\% \text{ per annum (cont.)}.$$

(b)

Use the Black–Scholes–Merton formula with continuous dividends and solve for σ from the call price:

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2), \quad d_{1,2} = \frac{\ln(S_0/K) + (r - q \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

Plugging $q = 2.992\%$ and $c = 78$ and solving numerically gives

$$\sigma \approx 24.68\% \text{ per annum}.$$

4

The implied volatilities for the call and put options with the same strike and maturity are different, which implies that their market prices violate put-call parity. The put's higher implied volatility means it is overpriced relative to the call.

The arbitrage strategy is to buy the call, sell the put, sell the stock, and invest in a risk-free bond with face value equal to the strike price $30 (i.e., invest \$30 e^{-r}$ now to receive $30 in 1 year, where \$r$ is the risk-free rate). This generates a positive cash inflow now and zero payoff at maturity, regardless of the stock price at maturity.

This arbitrage does not rely on the lognormal assumption underlying the Black-Scholes-Merton model. Put-call parity is a model-independent no-arbitrage condition for European options, derived solely from matching payoffs at maturity across the portfolio of the call, put, stock, and bond. It holds under any stock price distribution (including those with jumps or fat tails), as long as the options are European and trading frictions do not prevent holding the positions to maturity. The lognormal assumption is only relevant for computing Black-Scholes-Merton prices and implied volatilities, but the parity violation and resulting arbitrage are independent of it.